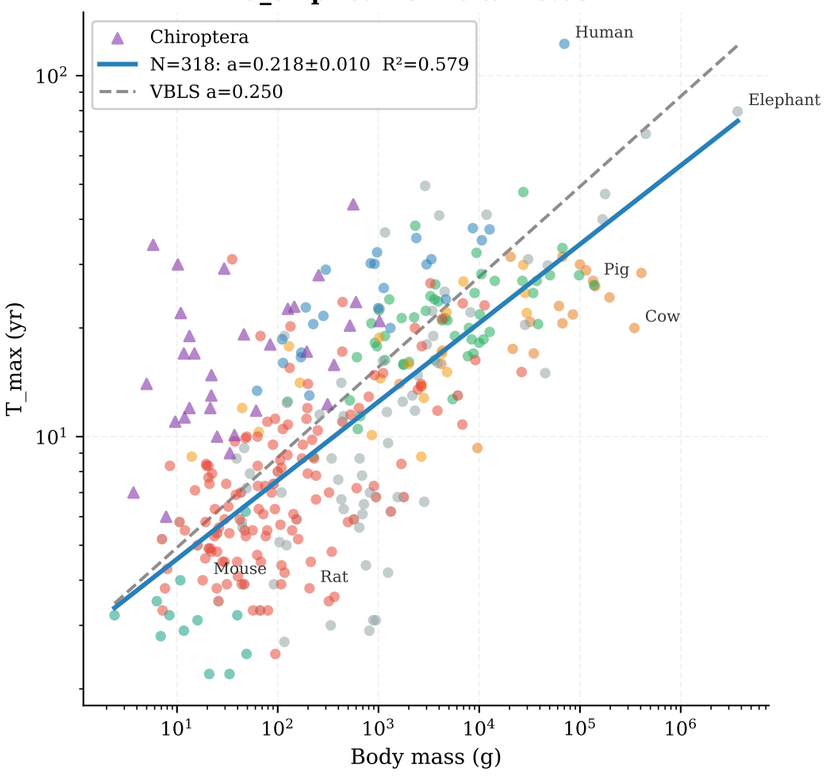
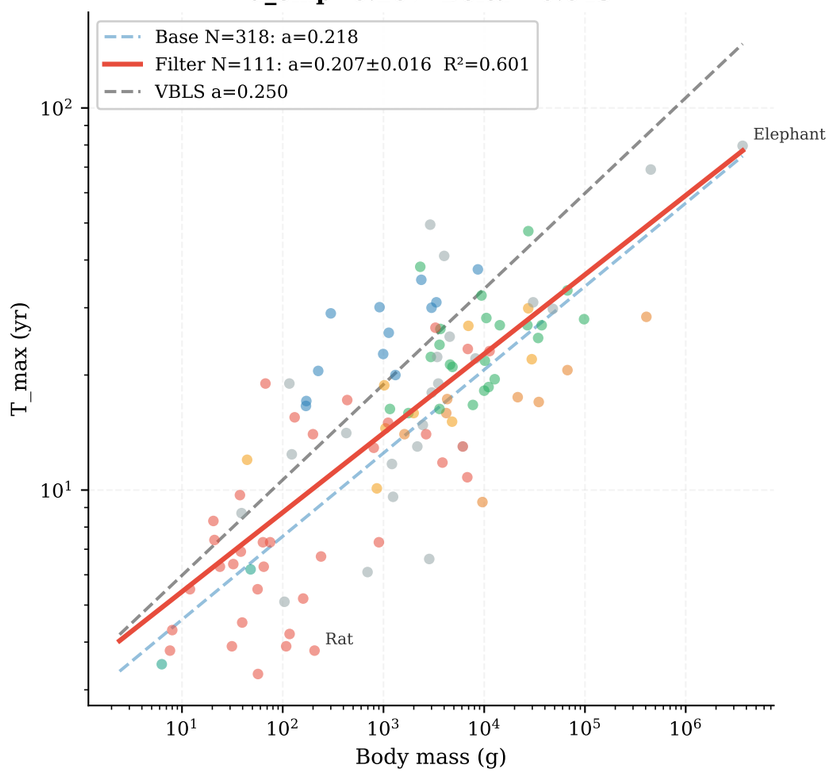
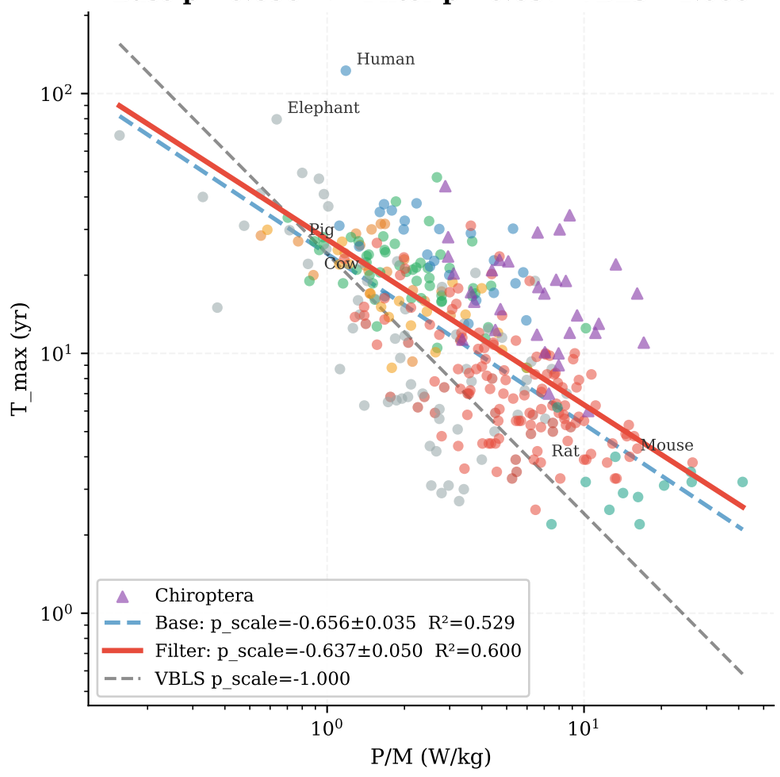
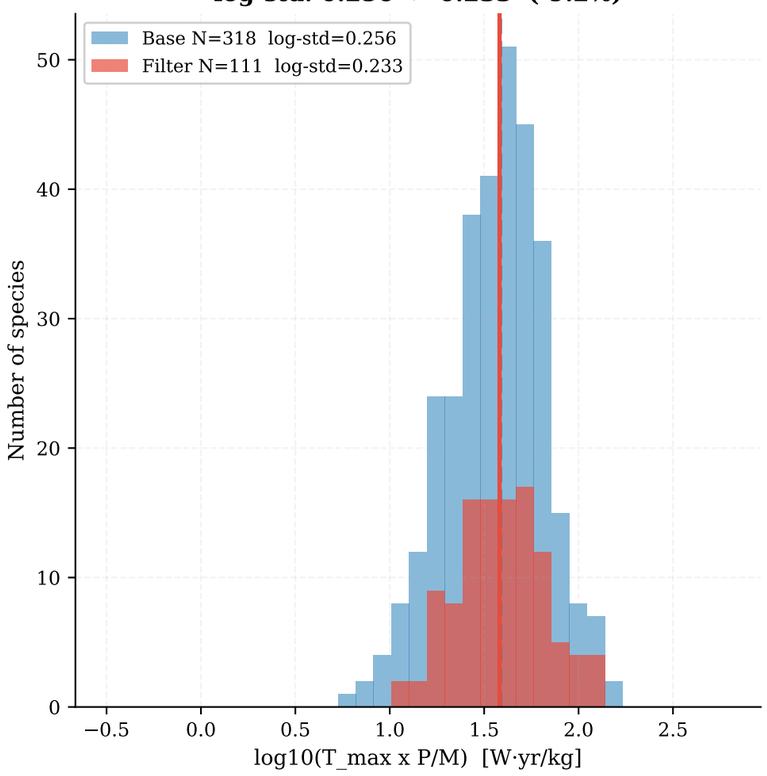
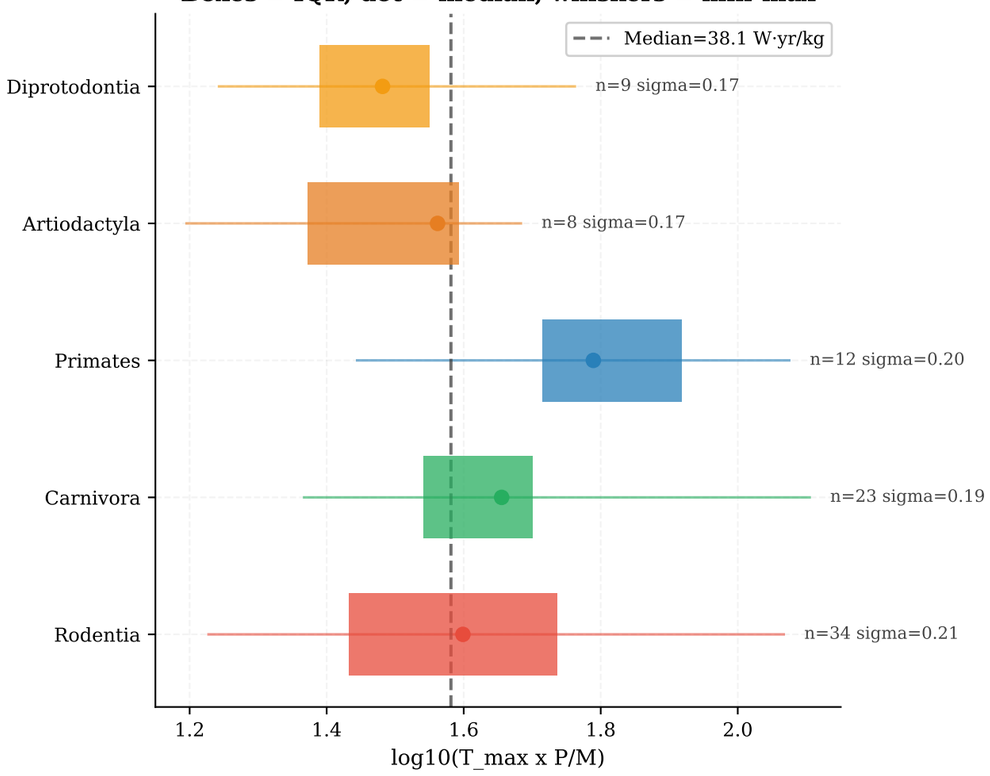



**F. Summary: base vs improved filtering**
**log-std — primary metric (not CV)**

| | VBLS | Base (N=318) | Filter (N=111) | Delta |
|---|---|---|---|---|
| a (T_max~M) | +0.250 | 0.218 ±0.010 | 0.207 ±0.016 | 0.043 vs 0.032 |
| beta (P/M~M) | -0.250 | -0.277 ±0.009 | -0.283 ±0.015 | 0.033 vs 0.027 |
| p (T_max~P/M) | -1.000 | -0.656 ±0.035 | -0.637 ±0.050 | 0.363 vs 0.344 |
| log-std (Rubner) | 0 | 0.2564 | 0.2327 | -0.0237 (9.2%) |
| 2sigma range | x1 | x3.26 | x2.92 | improved -0.34 |